

# **An Internship Report**

Of

## **Python Developer Internship**

Submitted

To

**School of Computer Science and Engineering**

**IILM University, Greater Noida, U.P.**

In partial fulfilment of the requirement of degree of

**B.Tech (Computer Science and Engineering)**



By

Roll Number of Student: 25SCS1003001040

**Name of Student: LAKSHYA TRIPATHI**

Section: 2CSE32

Semester: 3RD

Internship Duration: 06 August-06 August 2026 (1 Month)

## **CANDIDATE'S DECLARATION**

I, LAKSHYA TRIPATHI, do hereby declare that the internship report titled "Python Developer Internship" has been prepared by me to fulfil the academic requirement of the Bachelor of Technology programme in Computer Science and Engineering. The report documents the one-month internship completed with Codec Technologies from 06 August to 06 August 2026.

I affirm that this report is based on the internship learning material, project briefs and portal records supplied by me. All external references used for technical explanation have been acknowledged. This report has not been submitted to another institution for any degree or diploma requirement, and I accept responsibility for the originality of its contents.

Signature: \_\_\_\_\_

**Student Name: LAKSHYA TRIPATHI**

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**Section:**

**2CSE32**

**Semester: 3RD**

## ACKNOWLEDGEMENT

I take this opportunity to express my gratitude to everyone who supported me during the internship programme. Their guidance, encouragement and constructive advice helped me complete the learning activities and present the experience in an academic form.

I sincerely thank Codec Technologies for providing a structured Python Developer internship covering Python fundamentals, advanced programming concepts, data handling, APIs and industrial project work. The four-week sequence offered a practical way to connect programming concepts with application-oriented outcomes.

I am also thankful to IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous academic support and encouragement. Finally, I express my gratitude to my family for their motivation and blessings.

Signature: \_\_\_\_\_

**Student Name: LAKSHYA TRIPATHI**

**Roll No.: 25SCS1003001040**

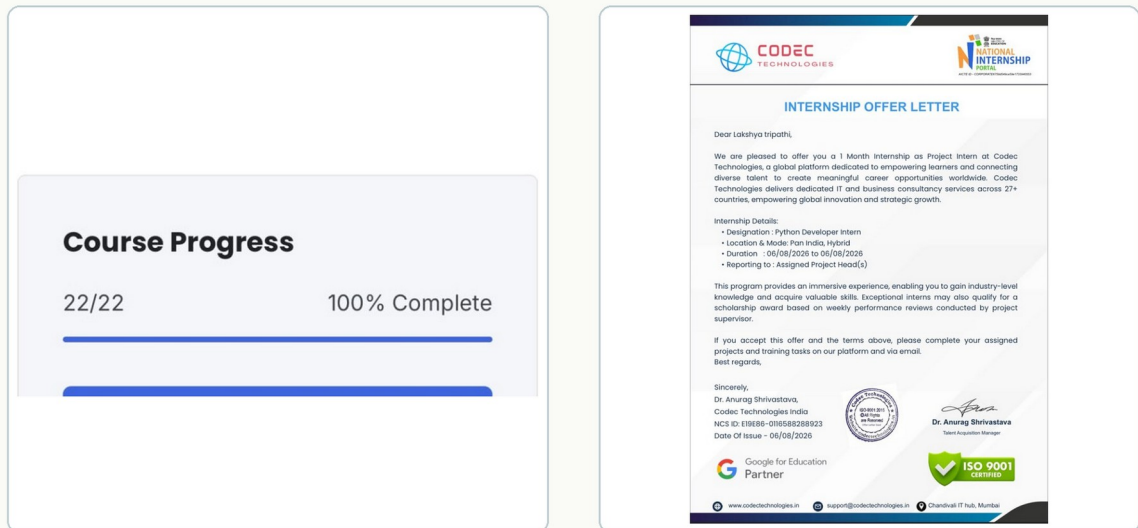
# TABLE OF CONTENTS

|                                              |    |
|----------------------------------------------|----|
| 1. Candidate's Declaration.....              | i  |
| 2. Acknowledgement.....                      | ii |
| 3. Internship Completion Evidence.....       | -  |
| 3.1 Portal Completion Evidence.....          | -  |
| 3.2 Internship Offer Letter.....             | -  |
| 3.3 Industrial Project Briefs.....           | -  |
| 4. Project Description.....                  | 1  |
| 4.1 Introduction.....                        | 1  |
| 4.2 Organization Profile.....                | 1  |
| 4.3 Problem Statement.....                   | 1  |
| 4.4 Project Objectives.....                  | 2  |
| 4.5 Scope of the Project.....                | 2  |
| 4.6 Technologies and Tools Used.....         | 3  |
| 4.7 System Architecture.....                 | 3  |
| 4.8 Methodology.....                         | 4  |
| 4.9 Expected Outcomes.....                   | 5  |
| 4.10 Completion and Communication Proof..... | 5  |
| 4.11 Challenges and Responses.....           | 6  |
| 4.12 Skills Acquired.....                    | 6  |
| 4.13 Conclusion and Future Scope.....        | 7  |
| 5. Bibliography/References.....              | 8  |

### 3. INTERNSHIP COMPLETION EVIDENCE

## Completion Evidence

Lakshya Tripathi - Python Developer Internship

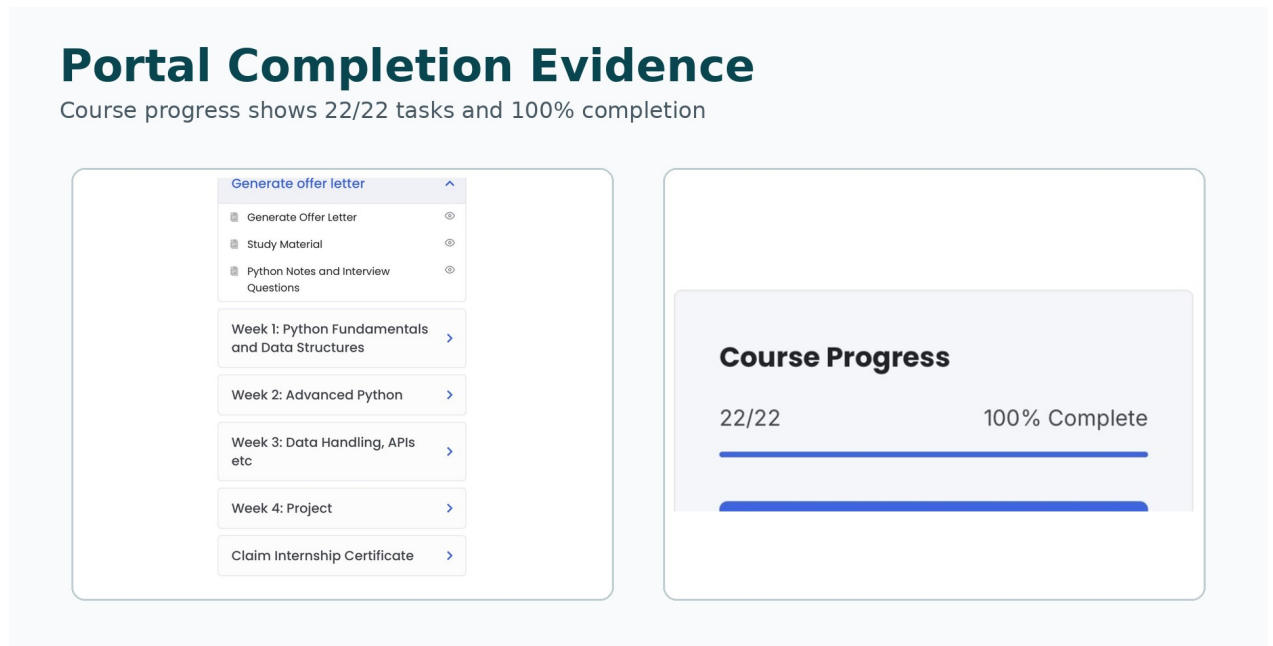


*Completion evidence supplied for Lakshya Tripathi.*

**Internship Role: Python Developer Intern**

The supplied portal evidence shows 22/22 tasks and 100% completion, while the offer letter confirms the Python Developer Intern designation.

### 3.1 PORTAL COMPLETION EVIDENCE



**Figure 3.1: Codec Technologies learning portal showing 100% course completion.**

*Source: User-supplied Codec Technologies portal screenshot captured 8 September 2026.*

The portal identifies the programme as a one-month Python Developer Internship and displays overall progress of 100%. It also lists learning areas for Python fundamentals, advanced Python, data handling and APIs, and the Week 4 project component. This record supports the completion status used throughout this report.

### 3.2 INTERNSHIP OFFER LETTER

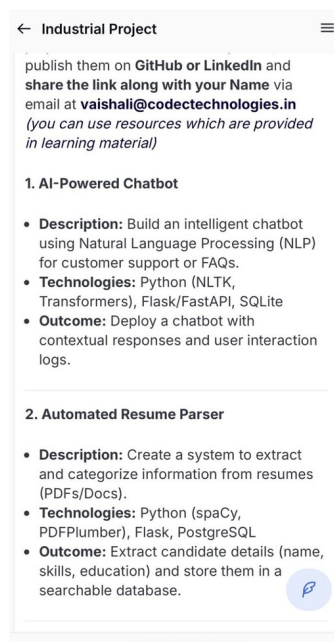


**Figure 3.2: Internship offer letter issued to Lakshya Tripathi by Codec Technologies.**

Source: User-supplied Codec Technologies offer letter dated 06 August 2026.

The offer letter appoints Lakshya Tripathi as a Python Developer Intern for the period from 06 August to 06 August 2026. It describes the internship as project-based, conducted in pan-India hybrid mode, and focused on industry-level knowledge, task completion and performance review.

### 3.3 INDUSTRIAL PROJECT BRIEFS



**Figure 3.3: Portal briefs for the AI-Powered Chatbot and Automated Resume Parser.**

*Source: User-supplied Codec Technologies industrial-project screenshot captured 8 September 2026.*

The portal asks the intern to select any two industrial projects and publish them through GitHub or LinkedIn. This report analyses the first two displayed briefs—an AI-powered support chatbot and an automated resume parser—as representative applications of Python, NLP, APIs and data storage. Repository links or measured implementation results were not supplied, so the briefs are treated as technical specifications rather than independently verified deployments.



## **4. PROJECT DESCRIPTION**

### **4.1 Introduction**

This report presents a detailed account of the one-month Python Developer internship completed with Codec Technologies from 06 August to 06 August 2026. The programme formed part of the student's practical learning in the Bachelor of Technology curriculum at IILM University, Greater Noida.

The internship followed a progressive learning sequence. It began with Python fundamentals, advanced to functions, object-oriented programming and exception handling, continued through data handling and API concepts, and concluded with industrial project briefs. This progression reflects the main stages involved in designing reliable Python applications.

### **4.2 Organization Profile**

Codec Technologies describes itself in the supplied offer letter as a global learning platform and an ISO-certified IT and business consultancy serving clients across more than 27 countries. Its internship model combines online learning material, staged technical topics and project-oriented assignments.

The portal-based format supports guided independent learning. Each stage introduces a focused topic, while the intern is expected to study the material, interpret project requirements, develop solutions, test the results and communicate completed work through professional channels.

### **4.3 Problem Statement**

A learner may understand Python syntax, data structures, functions and libraries as separate topics yet still find it difficult to combine them into a reliable application that accepts real input, processes data, connects to services and produces useful output.

The internship addressed this gap through a connected sequence of practical areas: writing clear program logic, organizing reusable modules, handling files and structured data, consuming APIs, managing errors and planning an integrated industrial project. Each stage added complexity while depending on knowledge gained earlier.

#### **4.4 Project Objectives**

- To apply Python syntax, variables, data types, operators and control structures correctly.
- To design reusable programs with functions, modules and object-oriented principles.
- To handle files, exceptions and common runtime errors in a controlled manner.
- To read, clean and transform CSV and JSON data using suitable Python tools.
- To understand database interaction and structured storage for application records.
- To consume web APIs, validate responses and process returned JSON data.
- To study NLP workflows for conversational support and document information extraction.
- To organize, document and share project work through a professional development workflow.

#### **4.5 Scope of the Project**

The scope covers the learning path visible in the portal: Python fundamentals, advanced Python, data handling, APIs and industrial project work. The technical analysis focuses on two displayed project specifications—an AI-powered chatbot and an automated resume parser—because together they connect NLP, web services, document processing and data storage.

The supplied evidence confirms overall course completion, the offer letter and portal completion evidence. It does not provide source-code repositories, deployment URLs, test reports or measured model performance. Those items are therefore described as recommended deliverables or future verification work, not as independently confirmed internship outputs.

## 4.6 Technologies and Tools Used

- Core Python: syntax, data types, loops, functions, modules and reusable program structure.
- Advanced Python: object-oriented programming, file handling and exception management.
- Data Handling: CSV, JSON, Pandas concepts, SQL and lightweight SQLite storage.
- API Development: HTTP requests, response validation, Flask or FastAPI service patterns.
- NLP and Parsing: NLTK or Transformers for chatbot logic; spaCy and PDFPlumber for resume extraction.
- Delivery: Git and GitHub or LinkedIn sharing, documentation, testing and deployment planning.

## 4.7 System Architecture

The general architecture implied by the curriculum and project briefs can be summarized as the following connected layers:

- Input Layer: text, resumes, forms, files or API requests enter the application through a defined interface.
- Validation Layer: input types, missing fields, file formats and unsafe values are checked before processing.
- Application Layer: Python functions and classes coordinate the required business or project logic.
- Processing Layer: data cleaning, NLP, parsing or classification transforms raw input into structured information.
- Persistence Layer: SQLite or PostgreSQL stores user interactions, extracted records and searchable metadata.
- Service and Output Layer: Flask or FastAPI returns responses, predictions or search results and records errors for review.

## 4.8 Methodology

The internship followed a milestone-based methodology combining concept study, guided practice, implementation, testing and project planning. The weekly portal structure created a clear progression from foundations to an industrial application, as summarized below.

| Milestone | Stage   | Learning Area        | Primary Learning                                 |
|-----------|---------|----------------------|--------------------------------------------------|
| 1         | Week 1  | Python Fundamentals  | Syntax, data types, operators and control flow   |
| 2         | Week 2  | Advanced Python      | Functions, OOP, files and exception handling     |
| 3         | Week 3A | Data Handling        | CSV, JSON, Pandas and structured storage         |
| 4         | Week 3B | APIs and Integration | HTTP requests, JSON responses and error handling |
| 5         | Week 4  | Industrial Projects  | NLP, document parsing, testing and sharing       |

**Note:** The portal confirms 100% course completion; the week labels summarize the visible curriculum sequence rather than separate dated submissions.

At each stage, the intern first studied the relevant concepts, converted them into small programs, checked expected behaviour and connected the results to the next topic. The final project stage required selecting two briefs, developing the solutions and preparing them for publication through GitHub or LinkedIn.

## **4.9 Expected Outcomes**

On completion of the internship, the documented or specified outcomes include:

- A stronger command of Python syntax, control flow, functions and reusable modules.
- Programs that use object-oriented design, file handling and structured exception management.
- The ability to process CSV and JSON data and organize records for database storage.
- API clients or lightweight endpoints that validate requests and return structured responses.
- An AI-powered chatbot specification using NLP for contextual support or frequently asked questions.
- An automated resume parser specification for extracting candidate details into searchable records.
- A disciplined development workflow connecting analysis, implementation, testing, documentation and sharing.

## **4.10 Completion and Communication Proof**

The Codec Technologies portal reproduced in Chapter 3 displays 100% completion for the one-month Python Developer Internship. The navigation also lists the learning sequence from Python fundamentals and advanced Python through data handling, APIs and the final project component.

The supplied offer letter identifies Lakshya Tripathi as a Python Developer Intern and records the internship period as 06 August 2026 to 06 August 2026. The industrial-project page instructs participants to publish two selected projects on GitHub or LinkedIn; however, no repository or post URL was supplied with the evidence reviewed for this report.

## 4.11 Challenges and Responses

The curriculum and project requirements reveal several common technical failure points. They can be addressed through disciplined implementation and testing responses:

| Challenge                      | Technical Response                                 | Intended Effect                    |
|--------------------------------|----------------------------------------------------|------------------------------------|
| Inconsistent or missing input  | Validate types, required fields and file formats   | Fewer preventable runtime failures |
| Complex program logic          | Separate work into functions, classes and modules  | Clearer testing and maintenance    |
| API requests fail              | Use timeouts, status checks and exception handling | More reliable service integration  |
| Unstructured text or documents | Preprocess, parse and store normalized records     | Consistent and searchable output   |

## 4.12 Skills Acquired

- Clear Python programming with appropriate data types, control flow, functions and modules.
- Object-oriented design, file operations and structured exception handling.
- Data cleaning and transformation using CSV, JSON, Pandas and SQL concepts.
- HTTP API integration, response checking and safe JSON processing.
- NLP and document-parsing concepts for chatbot and resume-processing workflows.
- Progressive problem solving, testing awareness, documentation and time management.

### 4.13 Conclusion and Future Scope

The four-week internship established a practical connection between the principal layers of Python development. Core syntax provided the foundation; functions and object-oriented programming improved structure; data handling introduced repeatable processing; APIs connected applications to external services; and the project briefs introduced NLP and document-processing use cases.

The most important outcome is the ability to view these topics as one development workflow rather than as isolated lessons. A useful Python application must accept validated input, apply maintainable logic, manage data safely, handle failures and communicate a clear result. The portal evidence confirms completion of the internship programme.

Recommended future learning includes:

- Using Git and GitHub consistently for version control, review and project history.
- Adding automated unit, integration and API testing with clear test coverage.
- Strengthening database design, authentication, privacy and secure file handling.
- Evaluating NLP quality with representative test data and measurable performance criteria.
- Packaging and deploying Python services with monitoring, logging and CI/CD practices.

The internship therefore provides a sound foundation for larger academic and portfolio projects. The next stage is to implement the selected briefs end to end, publish verifiable repositories and evaluate them with stronger testing, security and measurable technical quality.

## 5. BIBLIOGRAPHY/REFERENCES

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*Web resources accessed 8 September 2026. User-supplied internship evidence is reproduced in Chapter 3.*